

Fairness-Aware Recidivism Classification on COMPAS: Reproducing and Diagnosing Algorithmic Bias

Kosar Mazaheri

Digital Transformation Management

University of Bologna

Email: kosar.mazahery@studio.unibo.it

Abstract—The COMPAS recidivism risk tool was shown by ProPublica in 2016 to produce substantially higher false positive rates for Black defendants than for white defendants. We ask a narrower question than whether COMPAS is biased: whether a standard, off-the-shelf classifier trained on the same data independently reproduces that bias, and whether removing the protected attribute is sufficient to remove it. Training a tuned random forest on five screening-time features, we reach 0.684 accuracy—comparable to COMPAS’s own 0.661 on the same test defendants, and far below what would be needed for reliable individual prediction. Comparing the fairness of two predictors requires matching their *selection rates*, since a predictor that flags fewer people has a lower false positive rate for every group automatically; we show that neglecting this makes our model look fairer than COMPAS when in fact it is less fair. At a matched selection rate our model, despite never being shown COMPAS’s scores, imposes an identical false-positive burden on Black defendants while extending more benefit of the doubt to white ones, raising the false-positive disparity ratio from 1.81 to 2.19. We trace the mechanism to `priors_count`, which is both the dominant predictor and strongly correlated with race, and show that removing `race` from the feature set does not reduce the false positive rate for Black defendants at all. Finally, an intersectional analysis separates two properties that aggregate metrics conflate—the group flagged most *often* is not the group flagged most *unjustly*—and reveals that our model introduces a disparity with respect to sex roughly four times larger than COMPAS’s own.

Index Terms—Algorithmic fairness, recidivism prediction, COMPAS, proxy variables, machine learning.

1. Introduction

Risk assessment instruments are used in United States courts to inform decisions about bail, sentencing and parole. The most widely discussed of these is COMPAS, which assigns each defendant a risk decile from 1 to 10. In 2016 ProPublica published an analysis arguing that COMPAS was racially biased: among defendants who did not go on to re-offend, Black defendants were labelled high risk roughly twice as often as white defendants [1]. The tool’s vendor

replied that the instrument is calibrated within groups, meaning that a given score carries the same meaning regardless of race. Both claims turned out to be correct, and the apparent contradiction has since been shown to be mathematically unavoidable [2], [3].

This project does not attempt to settle that dispute. It asks a different and, we think, more tractable question. COMPAS is proprietary, and it is tempting to attribute its bias to the specific—and undisclosed—way its 137-item questionnaire is combined into a score. If that were the explanation, the remedy would be to build a better model. We therefore test the alternative hypothesis: that the disparity is a property of the *data* rather than of any particular algorithm, and that an ordinary classifier, built with standard tools and never shown COMPAS’s output, would reproduce it anyway.

Our contributions are as follows. First, we train and tune a conventional classifier on the ProPublica COMPAS data and evaluate it against both a naive baseline and COMPAS itself, establishing how little predictive headroom exists in this problem. Second, we show that a valid fairness comparison between two predictors requires holding their selection rates fixed, and that neglecting this inverts the conclusion. Third, at a matched selection rate we find that our model not only reproduces COMPAS’s directional bias but amplifies it. Fourth, we identify the mechanism: prior-arrest count is simultaneously the strongest predictor available and a strong proxy for race, so racial information reaches the model whether or not `race` is supplied as a feature; the simplest mitigation—removing the protected attribute—fails for exactly this reason. Fifth, an intersectional analysis separates two properties which aggregate metrics conflate: which group is flagged most often, and which group is flagged most unjustly. These turn out to be different groups. Finally, the same analysis exposes a disparity our model *introduces* rather than *inherits*: it relies on `sex` far more heavily than COMPAS does.

Group organization. This project was carried out individually. All stages of the work—problem definition, data cleaning, exploratory analysis, model development, the fairness audit and the writing of this paper—were performed by the author. The implementation is available as a single Google Colab notebook in the accompanying repository and executes end to end without errors.

2. Related Work

The original dispute. ProPublica’s *Machine Bias* investigation [1] released the dataset used here and reported that COMPAS’s errors were distributed unequally: Black defendants who did not re-offend were far more likely to have been labelled high risk, while white defendants who did re-offend were more likely to have been labelled low risk. Northpointe, the vendor, responded that the instrument satisfies predictive parity—a defendant assigned a given score has approximately the same probability of re-offending irrespective of race.

The impossibility results. Both positions are defensible because they appeal to different definitions of fairness. Chouldechova [2] showed that when the base rate of the outcome differs between groups, a classifier cannot simultaneously achieve predictive parity and equal false positive and false negative rates. Kleinberg et al. [3] proved a closely related impossibility for calibration and balance conditions. The practical consequence is that the choice of fairness criterion is a normative decision, not a technical one, and that reporting a single fairness number is insufficient. We therefore report several: selection rate, statistical parity difference, and group-conditional true and false positive rates.

Mitigation strategies. Existing methods intervene at three points in the pipeline. *Pre-processing* methods alter the training data; the best known is the reweighing scheme of Kamiran and Calders [4], which assigns instance weights so that the protected attribute and the label become statistically independent in the training distribution. *In-processing* methods add a fairness constraint to the learning objective. *Post-processing* methods adjust an already-trained model’s outputs; Hardt et al. [5] derive group-specific decision thresholds that equalise odds. A fourth approach, often called *fairness through unawareness*, simply omits the protected attribute from the feature set. It is the most intuitive strategy and the one most frequently proposed by non-specialists, and it is the one we evaluate here—not because we expect it to succeed, but because its failure is instructive.

Predictability of recidivism. Dressel and Farid [6] reported that COMPAS’s accuracy is matched both by a logistic regression on two features and by untrained human respondents recruited online. This finding frames our own results: the low accuracy we obtain is not a deficiency of our model but a property of the task.

3. Proposed Method

3.1. Data and preprocessing

We use the `compas-scores-two-years` dataset published by ProPublica [1], comprising 7,214 defendants screened in Broward County, Florida, described by 53 columns. We apply ProPublica’s published filters: the arrest and the COMPAS screening must fall within 30 days of each other; records with no matching COMPAS case are removed; ordinary traffic offences are removed; and records with a missing risk category are removed. This leaves 6,172

defendants. The target, `two_year_recid`, is close to balanced (45.5% positive), so accuracy is a meaningful headline metric and stratified splitting is appropriate.

3.2. Feature selection

Most of the 53 columns are inadmissible as model inputs, and we exclude them on three grounds. *Leakage*: fields such as `is_recid`, the re-offence records and the custody dates are recorded at or after the outcome being predicted, and a model given them would not be performing prediction at all. *Circularity*: COMPAS’s own scores are the object of the audit, so using them as inputs would amount to re-predicting COMPAS from COMPAS; we retain `decile_score` solely to construct the baseline. *Non-informative*: identifiers, names, dates and free-text charge descriptions carry no generalisable signal.

This leaves five screening-time attributes—that is, facts available at the moment the assessment is made: `age`, `priors_count`, `sex`, `c_charge_degree` and `race`. The last is retained not as an ordinary predictor but as the sensitive attribute required for the fairness analysis and for the mitigation experiment of Section 3.7. We do not prune further by hand: the tree-based models perform their own feature selection, and we report the resulting importances rather than imposing our own ranking.

3.3. Baseline: the COMPAS scores

Following ProPublica’s convention we binarise COMPAS’s decile score at 5, so that deciles 5–10 constitute a positive prediction. This threshold corresponds exactly to the union of the *Medium* and *High* risk categories, so the positive class should be read as “not low risk” rather than “high risk”—which is why it captures 44.2% of the test defendants.

3.4. Models

Categorical features are one-hot encoded and numeric features passed through, in a single `ColumnTransformer` wrapped with the classifier in a pipeline so that encoding is refitted within each cross-validation fold. We hold out 30% of the data as a test set (1,852 defendants), stratified on the target, and compare logistic regression, a decision tree and a random forest by five-fold cross-validation on the training set alone. The random forest is then tuned by grid search over the number of trees and the maximum depth. The test set is used exactly once, at the end.

3.5. Fairness metrics and the selection rate

Let A denote the protected attribute and $\hat{Y}$ the prediction. We report the *selection rate* $P(\hat{Y}=1)$, the statistical parity difference $SPD = P(\hat{Y}=1 \mid A=a) - P(\hat{Y}=1 \mid A=b)$, and the group-conditional true and false positive rates. The

false positive rate is the quantity of primary interest: a false positive means a defendant who did not re-offend was nonetheless labelled high risk, which is the error that carries a concrete cost for the individual.

Two conditions must hold for a comparison between two predictors to be meaningful. The first is that they be evaluated on the same individuals: we index COMPAS’s scores by the same test-set rows as our model’s predictions, so both are judged on the identical 1,852 defendants against the identical ground truth, and neither has seen those rows in training.

The second condition is less obvious and, we will show, decisive. The selection rate is not a property of a model but of the threshold applied to its output. A classifier ranks defendants by predicted probability; the threshold merely determines where that ranking is cut into “flag” and “do not flag”. A predictor that flags fewer people therefore produces fewer false positives for *every* group, and its FPR falls across the board irrespective of bias. In the limit, a predictor that flags nobody has a false positive rate of zero for all groups and is perfectly “fair” while being entirely useless. Comparing false positive rates at different selection rates thus measures a predictor’s willingness to flag as much as its bias. We therefore additionally evaluate our classifier at the probability threshold for which its selection rate equals COMPAS’s, and report the fairness metrics at that common operating point.

3.6. Proxy and intersectional analysis

To explain the results, we perform two further analyses. First, we isolate the top decile of `priors_count` and examine its racial composition, in order to test whether that feature encodes racial information. Second, we cross race, sex and charge degree and compute, within each cell, both the selection rate and the realised recidivism rate; their difference, which we call *over-flagging*, measures how far a group’s assigned risk exceeds its outcome. Cells with fewer than 50 defendants (20 in the test set) are excluded as uninterpretable.

3.7. Mitigation

We evaluate fairness through unawareness: the tuned model is retrained with `race` removed from the feature set, all other settings held fixed, and the fairness metrics recomputed at the same threshold.

4. Results

4.1. Predictive performance

All three model families cluster tightly, with five-fold cross-validated accuracies of 0.672 (logistic regression), 0.671 (decision tree) and 0.681 (random forest). The tuned random forest achieves 0.684 accuracy and 0.735 AUC on the held-out test set. Table 1 places this in context.

TABLE 1. TEST-SET ACCURACY IN CONTEXT

Predictor	Accuracy
Naive baseline (always predict majority)	0.545
COMPAS decile score (≥ 5)	0.661
Our tuned random forest	0.684

TABLE 2. FAIRNESS METRICS ON THE SAME 1,852 TEST DEFENDANTS

	COMPAS (≥ 5)	Model (thr. 0.5)	Model (matched)
Selection rate	0.442	0.373	0.442
Accuracy	0.661	0.684	0.684
SPD	0.238	0.330	0.323
FPR, Black defendants	0.411	0.337	0.411
FPR, white defendants	0.227	0.133	0.188
FPR ratio	1.81	2.53	2.19

Our model improves on the naive baseline by roughly 14 points and performs comparably to COMPAS—0.684 against 0.661, a difference of 41 defendants out of 1,852—while using five features against COMPAS’s 137-item questionnaire. The consistency across model families indicates that the ceiling is imposed by the data rather than by model capacity, in agreement with Dressel and Farid [6]. No model of this kind is accurate enough to justify confident individual decisions.

4.2. Why the selection rate must be matched

Table 2 reports the fairness metrics for COMPAS and for our model at two operating points. The middle column is the model at scikit-learn’s default probability threshold of 0.5. Read on its own, it appears to show that our model is the fairer of the two: its false positive rate is lower than COMPAS’s for *both* groups, 0.337 against 0.411 for Black defendants and 0.133 against 0.227 for white defendants.

This is an artefact of caution rather than of equity. As Section 3.5 argues, the two predictors are not operating at the same point: COMPAS flags 44.2% of the test set, our model only 37.3%. The lower false positive rates follow mechanically from flagging fewer people, and tell us nothing about the distribution of those errors across groups. Note that the SPD, which is a difference of selection rates and therefore less sensitive to this effect, already points the other way.

Lowering our model’s threshold to 0.461 brings its selection rate to exactly COMPAS’s 44.2%. At this common operating point the artefact disappears and the conclusion reverses. Our model imposes precisely the same false-positive burden on Black defendants as COMPAS—0.411 in both cases—while extending *more* benefit of the doubt to white defendants, 0.188 against 0.227. The disparity ratio grows from 1.81 to 2.19. At an equal willingness to flag, the model trained without any access to COMPAS’s scores is the *less* fair of the two.

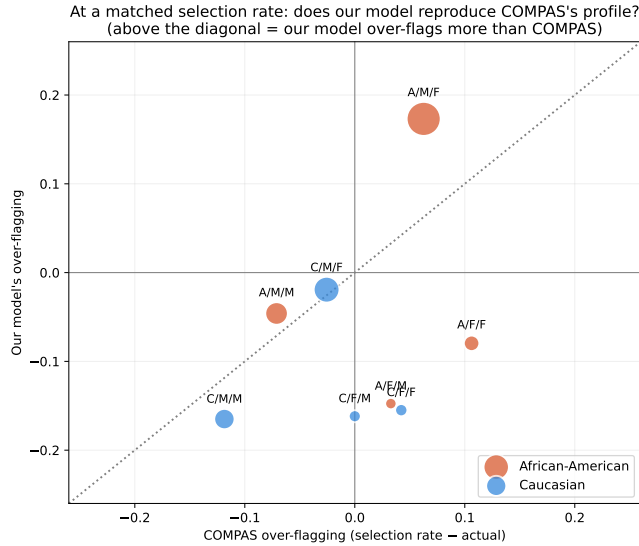


Figure 1. Over-flagging by intersectional cell, at a matched selection rate. The horizontal axis is COMPAS's over-flagging (selection rate minus realised recidivism); the vertical axis is our model's. Points above the dotted diagonal are groups our model over-flags *more* than COMPAS does. Bubble area is proportional to cell size; labels give race/sex/charge degree.

4.3. The model reproduces and amplifies the disparity

The intersectional cells, evaluated at the matched threshold and shown in Fig. 1, sharpen the same finding. Among men charged with a felony—the two largest cells, and therefore the most reliable—COMPAS over-flags Black defendants by 6.3 points relative to their realised recidivism rate, and under-flags white defendants by 2.6 points. Our model over-flags the former by 17.3 points, nearly three times COMPAS's excess, while leaving the latter essentially calibrated (−1.9 points). The selection-rate gap between these two groups accordingly widens from 25.4 points under COMPAS to 35.8 under our model.

The disagreement between the two predictors is not spread evenly across the sample. It is concentrated almost entirely on this single cell: the remaining male cells lie close to the line of identical behaviour. The female cells diverge for a different reason, examined in Section 4.7.

Our model was never given COMPAS's scores, its risk categories, or any derivative of them. That it independently recovers the same directional bias—and applies it more aggressively—indicates that the disparity is reproducible from the data alone, and is not an artefact of COMPAS's proprietary scoring.

4.4. The mechanism: priors as a proxy

The random forest's feature importances are dominated by `priors_count` (0.50) and `age` (0.31); `race` (0.07), `c_charge_degree` (0.07) and `sex` (0.05) follow. The dominant feature is also the one most entangled with race.

Black defendants constitute 51% of the sample but 75% of the top decile of prior counts; controlling for group size, 16.8% of Black defendants fall in that decile against 6.3% of white defendants. The decile is strongly predictive—its members re-offend at 75.7% against 41.6% for the remainder—which is precisely what makes it dangerous: the model has a strong incentive to rely on a feature that carries racial information.

4.5. Fairness through unawareness fails

Removing `race` and retraining leaves accuracy essentially unchanged (0.684 to 0.689) and reduces SPD from 0.330 to 0.272. Read carelessly, this looks like partial success. The disaggregated rates show it is not. The false positive rate for Black defendants is 0.337 with `race` and 0.339 without it: the disadvantaged group is not helped *at all*. The gap narrows only because the false positive rate for white defendants *rises*, from 0.133 to 0.173. The intervention does not reduce harm; it redistributes it. This is the predicted consequence of the proxy structure: with `priors_count` still present, the model reconstructs the racial signal it was denied. (Both configurations are evaluated at the same threshold, so this comparison is not subject to the selection-rate confound of Section 3.5.)

4.6. Flagged most often is not flagged most unjustly

We turn now to COMPAS's own scores across the full cleaned dataset, where the intersectional cells are largest. The group flagged most often is Black men charged with a felony, labelled high risk 62.7% of the time. The group flagged most *unjustly*—the largest excess of assigned risk over realised outcome—is a different one: Black women charged with a felony, who re-offend at 41.4% but are flagged 56.1% of the time, an excess of 14.7 points. The comparison that isolates the racial effect most cleanly is this: white men charged with a felony re-offend at a *higher* rate (43.3%) than Black women charged with a felony, yet are flagged *less* often (37.8%). Race is sufficient to reverse the effect of sex. Aggregate metrics cannot express this distinction; only the disaggregated view separates being flagged frequently from being flagged unfairly.

4.7. An introduced disparity: sex

Because `sex` is also a protected attribute, we audit it on the same terms and at the same matched operating point. The result is not a repetition of the racial finding but a distinct one, and it explains the position of the female cells in Fig. 1. COMPAS separates the sexes only weakly: it flags 45.6% of men and 38.3% of women (SPD = 0.072), with false positive rates of 0.308 and 0.266—close to parity. Our model flags 49.7% of men and 21.4% of women (SPD = 0.283), with false positive rates of 0.329 and 0.107. Its sex disparity is therefore roughly four times COMPAS's, and its false positive rate for men is three times that for women.

Part of this tracks a real difference in outcomes—men in the test set re-offend at 48.0% against 35.3% for women—but the 28-point gap in selection rates is more than double the 13-point gap in outcomes. Our model has learned to rely on `sex` considerably more heavily than COMPAS does, and in doing so *introduces* a disparity that the instrument under audit does not exhibit. This is a caution against assuming that a self-built model is safer than a proprietary one merely because its features are visible: knowing which attributes a model uses says nothing about how hard it leans on them. We report this disparity but do not mitigate it, and note the omission as a limitation.

5. Conclusions

We set out to test whether COMPAS’s racial disparity is a property of its proprietary algorithm or of the data it is trained on. The evidence points to the latter. A conventional random forest, given five ordinary features and never shown COMPAS’s output, reaches comparable accuracy and reproduces the same directional bias—in amplified form.

Establishing this required care. At its default threshold our model appeared *fairer* than COMPAS, with lower false positive rates for both racial groups. That appearance dissolved once the two predictors were compared at the same selection rate: the model had simply been flagging fewer people. At a matched operating point it imposes an identical false-positive burden on Black defendants while treating white defendants more leniently, and the disparity ratio rises from 1.81 to 2.19. An aggregate improvement had concealed a widening gap beneath it—which is, in miniature, the argument of the paper.

The mechanism is proxy encoding. Prior-arrest count is the strongest predictor available and is strongly correlated with race, so the model acquires racial information regardless of whether `race` appears in the feature set. Removing the protected attribute therefore fails, and fails instructively: it leaves the false positive rate for Black defendants untouched and narrows the gap only by worsening outcomes for white defendants. Any serious mitigation must address the correlated features, not merely the protected one; reweighing [4] and equalised-odds post-processing [5] are the natural next steps.

A final result cuts against a comfortable assumption. Our model does not merely inherit COMPAS’s biases; with respect to sex it introduces a disparity four times larger than the instrument it was auditing. Building one’s own model, with visible features and open code, is not by itself a safeguard.

Two caveats bound all of the above. Prior counts and charge records measure *arrests*, which reflect policing and enforcement patterns as much as offending, and the target label inherits the same defect, since re-arrest is only a proxy for re-offence. Every disparity reported here therefore describes the criminal-justice record rather than the people in it. And intersectional cells become small quickly, so our claims are restricted to the well-populated ones. Within

those bounds, the conclusion stands: the bias survives the removal of the attribute that names it.

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